Q 1-List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

Here are 5 common hardware issues that can cause network problems in a typical network:

(1) Faulty or Damaged Ethernet Cables: Frayed wires, internal breaks, or broken retention clips on RJ45 connectors can cause intermittent connectivity, high packet loss, or a complete loss of link.

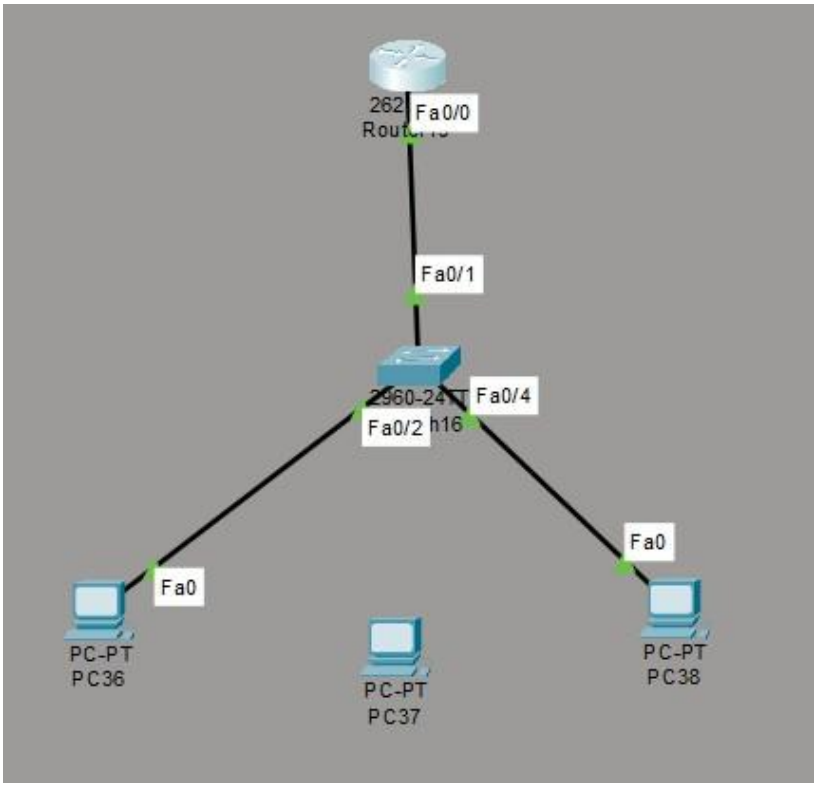
(2)Overheating Network Hardware: Inadequate ventilation or aging cooling fans in routers, switches, or modems can lead to thermal throttling, random reboots, and dropped packets.

(3)Failing Network Interface Card (NIC): A malfunctioning or poorly seated NIC on a computer or server can result in frequent disconnections or an inability to acquire an IP address.

(4)Unstable Power Supply: Power surges, voltage fluctuations, or a failing power adapter can cause networking devices to reset unexpectedly, disrupting active connections.

(5)Damaged or Dirty Switch/Router Ports: Physical wear, bent pins, or dust buildup inside Ethernet ports can prevent proper physical contact, often resulting in link negotiation failures (such as dropping from 1 Gbps to 100 Mbps).

Q 2-Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.

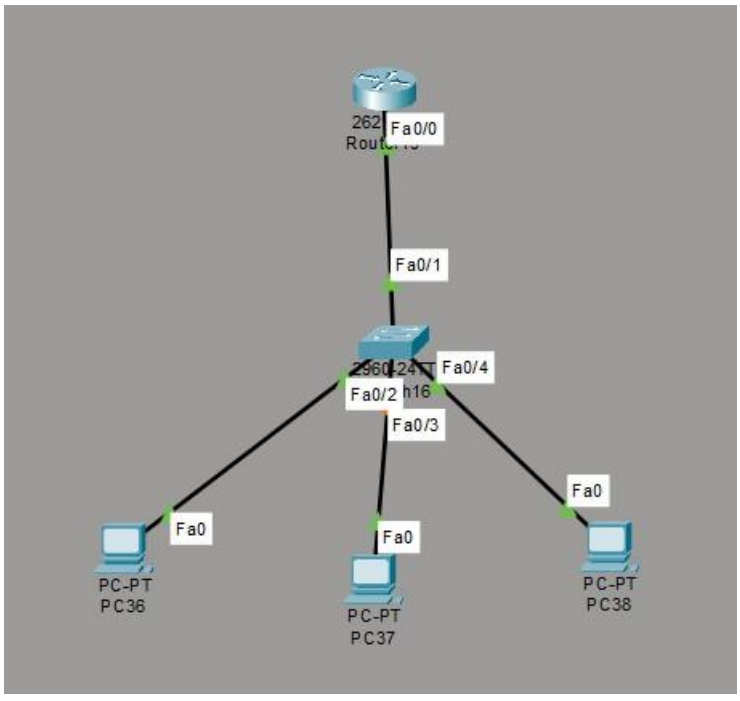



```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Router#
```



Troubleshooting steps:

1. Checked the physical topology.
2. Observed that PC1's switch connection was down.
3. Sent a ping from PC0 to PC1.
4. Ping failed with Request timed out.
5. Tested PC2 and confirmed that the rest of the network was working.
6. Used Packet Tracer Simulation Mode to observe the failed ICMP communication.
7. Reconnected PC1's cable.
8. Tested the connection again.
9. Ping succeeded.

Fault: PC1 Ethernet cable was intentionally disconnected.

Symptom: PC1 could not communicate with other devices.

**Q 3- Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.**

Step-by-Step Troubleshooting Checklist

No.	Troubleshooting Step	Action to Perform	Expected Result
1	Check cables and connections	Ensure the router's power cable and Ethernet cables are connected properly.	All cables are secure and undamaged.
2	Check router status lights	Observe the Power, Internet/WAN, and Wi-Fi indicator lights.	Lights show normal operation without unexpected errors.
3	Restart the router	Turn off the router, wait 30 seconds, and turn it on again.	The router reconnects and internet speed improves if the issue was temporary.
4	Check Wi-Fi signal strength	Move closer to the router and check the signal strength on your device.	A stronger signal provides a more stable connection.

5	Disconnect unused devices	Disconnect devices that are downloading files or streaming videos unnecessarily.	Less network congestion and improved available bandwidth.
6	Run an internet speed test	Use Speedtest by Ookla to measure download and upload speeds.	Results are compared with the speed promised by the ISP.
7	Test connectivity using ping	Open Command Prompt and run ping 8.8.8.8.	Replies indicate connectivity; high latency or packet loss may indicate a network problem.
8	Check IP configuration	Run ipconfig in Command Prompt and check the IPv4 address, default gateway, and DNS settings.	The device has a valid IP address and correct network configuration.
9	Update network drivers	Open Device Manager, locate the Wi-Fi adapter, and check for driver updates.	The wireless adapter operates correctly with compatible drivers.
10	Check router placement	Place the router in a central, elevated location away from thick walls and electronic interference.	Better Wi-Fi coverage and fewer signal interruptions.
11	Check background applications	Open Task Manager and check whether downloads, updates, or applications are consuming network bandwidth.	Unnecessary bandwidth usage is reduced.
12	Contact the ISP	If the issue continues, check for service outages or ask the ISP to test the connection.	The ISP identifies possible line faults, service issues, or speed limitations.

Verification Commands

Open Command Prompt (CMD) on your Windows computer and execute:

ipconfig

ping 8.8.8.8

ping google.com

- **ipconfig — displays IP configuration details.**
- **ping 8.8.8.8 — checks connectivity to a public IP address.**
- **ping google.com — checks connectivity using a domain name, also testing DNS resolution.**

Q 4-You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

Troubleshooting Steps

If **WhatsApp Web works but Instagram and YouTube don't**, I would perform these first 5 checks:

1. **Check Internet connectivity** ○ Open Command Prompt and run:

   ping 8.8.8.8

   - If it works, the laptop has basic Internet connectivity.
2. **Test the websites in another browser** ○ Try Instagram and YouTube in Chrome, Edge, or Firefox.
   - If they work in another browser, the problem is likely **browser/software-related**.
3. **Check DNS resolution** ○ Run:

   nslookup youtube.com nslookup instagram.com

   - If DNS fails, investigate **DNS/network configuration**.
4. **Check firewall, proxy, or security software** ○ Check whether a Windows firewall rule, antivirus, VPN, or proxy is blocking Instagram/YouTube.
   - In Windows, check **Settings → Network & Internet → Proxy**.
5. **Test another device on the same Wi-Fi** ○ Try Instagram and YouTube from a phone connected to the same Wi-Fi.
   - **Works on phone but not laptop:** likely laptop/software/configuration issue. ○ **Fails on all devices:** likely router, DNS, ISP, or network configuration issue.